

Composio App Intelligence Research

Take-Home Assignment Submission Summary

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1. Live Deliverable Links

Interactive Case Study & Live Dashboard URL:

<https://ishaanpgujaran.github.io/composio-product-research-agent/>

(Redirects automatically to static subpage site/index.html)

Source Code GitHub Repository:

<https://github.com/ishaanpgujaran/composio-product-research-agent>

2. Pipeline & Methodology Overview

This project implements an autonomous pipeline designed to audit applications for AI-agent readiness. Instead of manually inspecting documentation, a parallelized research agent searches developer resources, downloads docs, extracts auth methods, self-serve developer access, and MCP server metadata into structured schemas verified via Pydantic v2.

To ensure rigor, a verification sampler selects a stratified group of 20 apps. The human-in-the-loop CLI compares first-pass outputs with ground truth, mapping failures into an error taxonomy. The pipeline successfully analyzed the complete set of 100 applications using a custom multi-model rate-limit cascade that bypasses daily free-tier quotas. The dataset achieved an accuracy of 88.8% on a 20-app verification sample.

3. Setup & Execution Instructions

Follow these steps to run the pipeline locally:

```
# 1. Setup virtual environment
python3 -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt

# 2. Configure credentials in env file
cp .env.example .env # Edit .env to add GEMINI_API_KEY

# 3. Run parallel research operator batch
python -m src.batch --config data/apps.json --output data/pass_1.json

# 4. Generate stratified sample & run interactive verifier
python -m src.verify --sample
python -m src.verify --run
```

```
# 5. Regenerate analytical graphs and serve dashboard locally
python -m src.analyze
python3 -m http.server 8000 --directory site/
```